

Quantitative analysis of Sb donors in Si by photoluminescence due to bound exciton two-electron transition

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ABSTRACT

Bound exciton (BE) emission line from Sb donors in Si coincides spectrally with that from P donors, preventing their separation by conventional photoluminescence (PL) spectroscopy. In contrast, the energies of bound exciton two-electron (BE2e) transitions in which the final state is an excited impurity are strongly impurity dependent and allow clear discrimination. We demonstrate quantitative determination of Sb donors in Si using the intensity ratio of the BE2e transition to free exciton emission. Sb concentrations from 1×10^{12} to $5 \times 10^{15} \text{ cm}^{-3}$ are measured without interference from P donors, establishing BE2e PL as a practical method for Sb quantification.

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The rapid growth in demand for power devices, such as insulated gate bipolar transistors (IGBTs), has accelerated the need for large-scale production of high-resistivity n-type Si wafers.^{1–3} Although float-zone (FZ) crystals have been widely used for this purpose, their maximum diameter of 200 mm presents a significant limitation. Magnetic field-applied Czochralski (MCZ) growth enables the fabrication of high-quality 300 mm Si crystals; however, the low crystal yield of P, the most commonly used n-type dopant impurity, remains a critical issue due to its large segregation coefficient. Sb-doped MCZ crystals have attracted increasing interest^{4,5} because their high evaporation rate^{6–8} effectively compensates for segregation, improving crystal yield.^{9–12} For high-resistivity crystals, precise control of not only intentional dopants but also residual donor and acceptor impurities is essential, making accurate quantitative impurity analysis indispensable.^{13,14}

Photoluminescence (PL) spectroscopy is a standard method for impurity quantification in Si, using the intensity ratio of impurity-bound exciton (BE) line to free exciton (FE) line as a measure of the impurity concentration.^{13,15–20} This method has been standardized by JIS²¹ and SEMI.²² However, Sb BE lines coincide spectrally with P BE lines within experimental uncertainty,^{23–26} making their separation impossible. To address this issue, we previously developed a method to analytically separate the Sb and P components by exploiting

differences in the intensity ratios between the no-phonon (NP) and transverse optical (TO) phonon lines of the two impurities.²⁷ Nevertheless, this approach is indirect and inevitably involves sources of uncertainty.

In this work, we focus on BE two-electron (BE2e) transitions, in which part of the BE recombination energy excites the donor electron to a higher energy state.^{28,29} The energy separation between the ground and excited states depends strongly on the impurity species and is significantly larger than differences in BE recombination energies, enabling unambiguous donor identification. Although BE2e transitions have been reported previously,^{26,29–32} their weak emission intensity in the longer-wavelength region has limited their practical application. Recent advances in PL instrumentation now allow reliable detection of these transitions, and we demonstrate a direct and accurate method for quantifying Sb impurities in Si without interference from the presence of P donors.

The BE2e transitions were first observed in Si by Dean *et al.*²⁹ The process for a neutral donor BE is schematically illustrated in Fig. 1. A portion of the BE recombination energy is used to excite the donor electron from the ground state to excited states, which are s-like and therefore cannot be observed by far-infrared absorption spectroscopy.^{29–31,33} The recombination energy of the BE, $h\nu_{\text{BE}}$, and that of BE2e transition, $h\nu_{\text{BE2e}}$, are given by

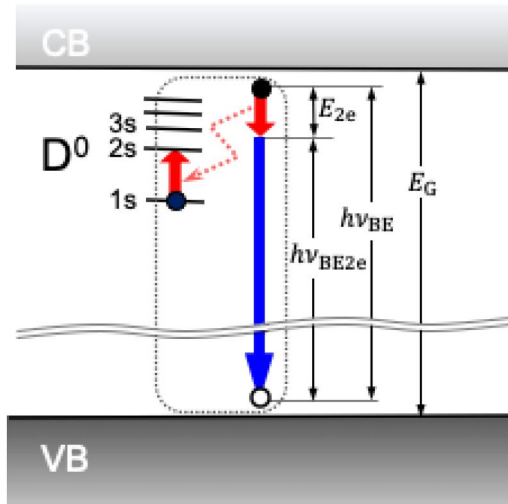


FIG. 1. Schematic illustration of BE2e transition in neutral donor. Part of BE recombination energy is used to excite donor electron from ground state to excited state (1s→2s in this case).

$$h\nu_{BE} = E_G - E_{FX} - E_{BX} - \hbar\omega, \quad (1)$$

$$h\nu_{BE2e} = E_G - E_{FX} - E_{BX} - E_{2e} - \hbar\omega, \quad (2)$$

where E_G is the bandgap energy, E_{FX} the formation energy of FE, E_{BX} the exciton binding energies to an impurity, $\hbar\omega$ the energy of emitted phonon for momentum conservation, and E_{2e} the difference between the ground and excited states, referred to as the displacement energy. Similar transitions occur for neutral acceptors, termed BE two-hole (BE2h) transitions with the displacement energy E_{2h} . For shallow donor and acceptor impurities, the difference in E_{BX} among impurities is typically less than 1 meV,^{16,20,23,24,26,31} whereas the difference in E_{2e} or E_{2h} is approximately one order of magnitude larger, ranging from a few meV to more than 10 meV.^{26,29–33} This large difference enables straightforward spectroscopic discrimination of impurity species.

Samples used in this study are Sb-doped MCZ Si, Sb- and P-codoped MCZ Si, and P-doped FZ Si (JIS standard sample:²¹ P-2) with the resistivity ranging from 39 to 751, 68, and 49 Ω cm, respectively, as listed in Table I; the net donor concentration calculated from the resistivity is also shown. The MCZ samples were cut out from high-quality 300 mm wafers for IGBT fabrications, and their

resistivity range covers the entire range used in current IGBT manufacturing. The Sb concentrations of samples S1–S4 were measured by SIMS. According to the analytical company, the detection limit of SIMS is $1 \times 10^{13} \text{ cm}^{-3}$ and the error is within approximately 40%. The Sb concentration of sample S5 was estimated to be $7.7 \times 10^{12} \text{ cm}^{-3}$ within 30% error by the standardized PL method after analytical spectral separation between Sb and P components.²⁷

PL measurements were performed at 4.2 K under conditions basically following JIS H 615:2021.²¹ The samples were immersed in liquid He in a quartz Dewar. A diode-pumped solid-state laser with a 532 nm line was used as the excitation source. The beam diameter was approximately 1 mm, and the irradiation power on the sample surface was set to 14 mW, corresponding to a power density of 1.8 W cm^{-2} , comparable to the JIS-defined range ($1.0\text{--}2.0 \text{ W cm}^{-2}$). Emission from the sample was dispersed by a spectrograph ($f=0.32 \text{ m}$, $F=4.9$) with a $600 \text{ groove mm}^{-1}$ grating blazed at $1.2 \mu\text{m}$, and detected using an InGaAs photodiode array cooled at -100°C . The spectral response of the measurement system was nearly flat over the photon-energy range of interest, and no spectral correction was applied. The spectral resolution was set to 0.26 nm (approximately 0.32 meV), and the vertical axis of the spectra is expressed to be proportional to the number of photons per unit energy interval.

Figure 2 compares PL spectra from samples P-2, S2, and C2, showing the NP and TO-phonon regions of the BE recombination and the TO-phonon region of the BE2e transition. The intensities are normalized to the peak intensity of the TO-phonon sideband of the BE line. The symbols assigned to the emission lines follow Refs. 15 and 16, with the impurity species indicated by the prefix (I for FE emission), phonons by the subscript, and the transition type indicated in parentheses; b_n is the n th bound multiple exciton complex (BMEC), and b_n' is the β series of BMEC.³¹ For BE2e transitions, the final state is specified; for example, “BE-2s” denotes excitation of the donor electron from the 1s ground state to the 2s excited state. As has been pointed out, the final states are even-parity states; 2s, 3s, ... for donors and $2\Gamma_8^+$, $3\Gamma_8^+$, ... for acceptors.^{29,30,32,33} To our knowledge, the lines marked X_P and X_{Sb} have not been reported previously. The fact that their intensities correlate with those of the principal BE2e lines allows us to suggest that they are attributable to BE2e transitions. The X_P and X_{Sb} lines do not overlap with either the principal lines $P_{TO}(\text{BE-2s})$ or $Sb_{TO}(\text{BE-2s})$, and their intensity is less than 1/10 of the principal lines. Therefore, they do not affect the following discussion.

The three spectra are nearly identical in the NP and TO-phonon regions of the BE emission, reflecting the fact that the BE recombination energies of Sb and P are identical within experimental

TABLE I. Specification of samples used in this study.

Sample no.	Dopant	Resistivity (Ω cm)	Donor conc. (cm^{-3})	Sb conc. by SIMS (cm^{-3})
S1	Sb	39	1.1×10^{14}	1.2×10^{14}
S2	Sb	81	5.3×10^{13}	6.0×10^{13}
S3	Sb	243	1.7×10^{13}	2.3×10^{13}
S4	Sb	262	1.6×10^{13}	2.2×10^{13}
S5	Sb	751	5.5×10^{12}	...
C2	Sb, P	68	6.3×10^{13}	3.0×10^{13}
P-2	P	49	8.8×10^{13}	...

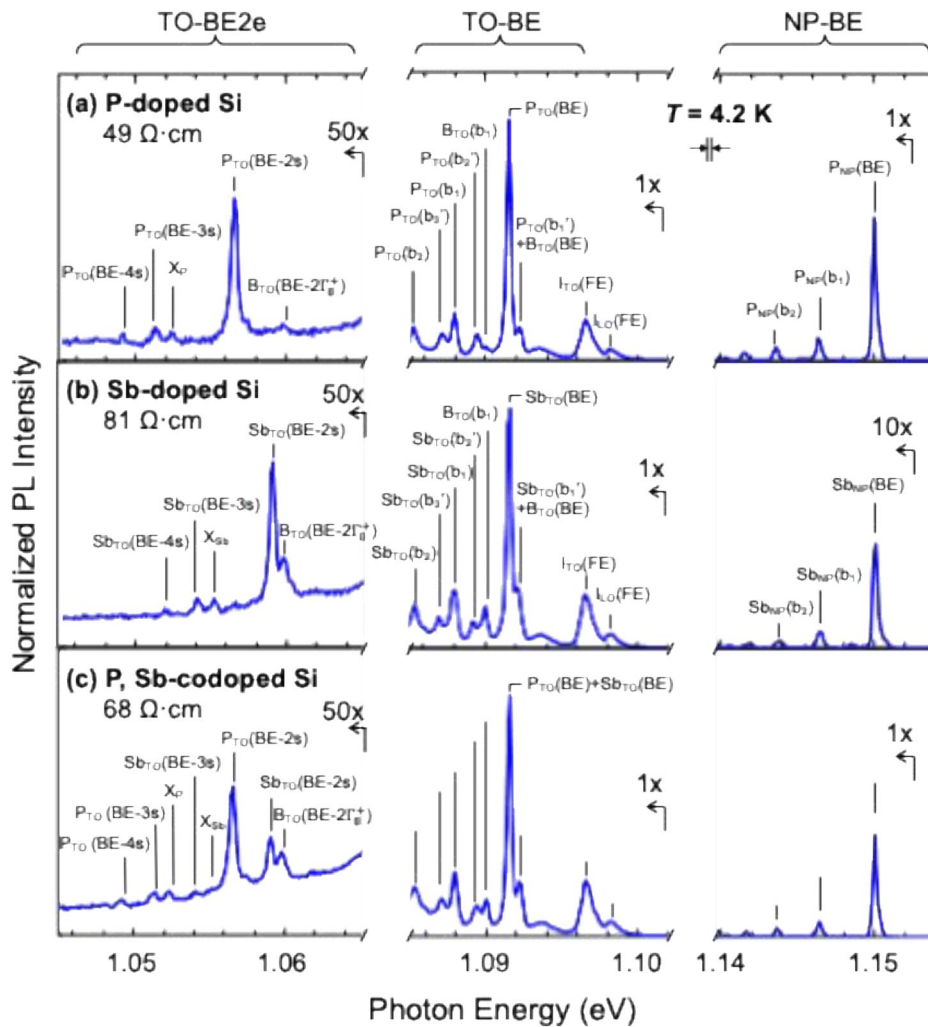


FIG. 2. PL spectra at 4.2 K of (a) P-doped Si (P-2), (b) Sb-doped Si (S2), and (c) P- and Sb-codoped Si (C2) with resistivity of 49, 81, and 68 $\Omega \cdot \text{cm}$, respectively. PL intensity is normalized to the peak intensity of the TO-phonon sideband of the BE line; symbol “10x” denotes amplitude factor. The three spectra are nearly identical in the NP and TO-phonon regions of the BE emission (right and center), whereas they differ significantly in the TO-phonon region of the BE2e transition (left), where Sb-, P-, and B-related spectral components are clearly separated and distinguishable.

uncertainty. The fact that the NP lines are very weak in sample S2 is the only characteristic distinguishing Sb from P.^{26,27,29} Small differences are also observed in the relative intensities of the B BE lines [$B_{\text{TO}}(\text{BE})$, $B_{\text{TO}}(b_1)$], with sample P-2 showing lower intensities than samples S2 and C2, reflecting its lower residual B concentration, and are unrelated to the P and Sb spectral features. In contrast, the PL spectra differ significantly among the three samples in the TO-phonon region of the BE2e transition, as shown in Fig. 2. Principal lines $Sb_{\text{TO}}(\text{BE-2s})$ and $P_{\text{TO}}(\text{BE-2s})$, as well as an additional line $B_{\text{TO}}(\text{BE-2}\Gamma_8^+)$, are clearly separated and distinguishable. The BE2e lines are also observed in the NP region; however the $Sb_{\text{NP}}(\text{BE-2s})$ line is extremely weak and below our detection limit, as can be expected from the NP/TO ratio of the Sb BE line, so we focused on the TO-phonon region.

We next investigate the dependence of the BE2e transition on the Sb concentration. Figure 3 shows the TO-phonon region of the BE2e transition for samples S1–S5. Principal $Sb_{\text{TO}}(\text{BE-2s})$ line, higher excited state lines $Sb_{\text{TO}}(\text{BE-3s})$ and $Sb_{\text{TO}}(\text{BE-4s})$, and residual P- and B-related lines $P_{\text{TO}}(\text{BE-2s})$ and $B_{\text{TO}}(\text{BE-2}\Gamma_8^+)$ are observed. It is

noteworthy that the P component was observed in all the samples in Fig. 3. The standardized PL method required indirect analytical separation of the P and Sb components. In the present case, we only needed to focus on the Sb component, without worrying about interference from the P component.

Following the procedure of the standardized PL method,²¹ we examine the dependence of the intensity ratio $Sb_{\text{TO}}(\text{BE-2s})/I_{\text{TO}}(\text{FE})$ on the Sb concentration. The intensity of the $Sb_{\text{TO}}(\text{BE-2s})$ line is determined by a baseline connecting the background areas on either side of the peak: an example of a baseline is shown in the spectrum of sample S5. The ratio is plotted against the Sb concentration, N_{Sb} , determined by SIMS for samples S1–S4 and by the standardized PL method after analytical spectral separation for sample S5 in Fig. 4. The error in the ratio arises from the noise level around the line of interest and the uncertainty of the baseline, and is within approximately 20%. This is consistent with the result of the round-robin test for the intensity ratio of BE to FE lines in the standardized PL method.^{16,19} The data for samples S1–S4 are well fitted by a line with a slope of 1, yielding

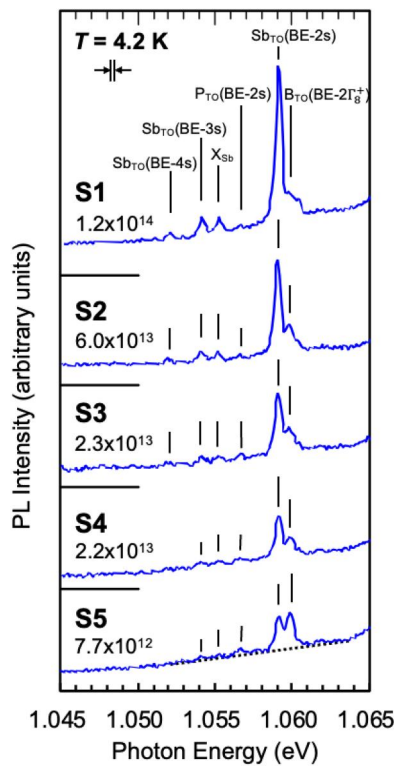


FIG. 3. PL spectra at 4.2 K of the TO-phonon region of BE2e transition for Sb-doped Si (samples S1–S5). Numbers attached to each sample indicate Sb concentrations in units of cm^{-3} . Baseline used to determine the intensity of the BE2e line is shown by dotted line in sample S5.

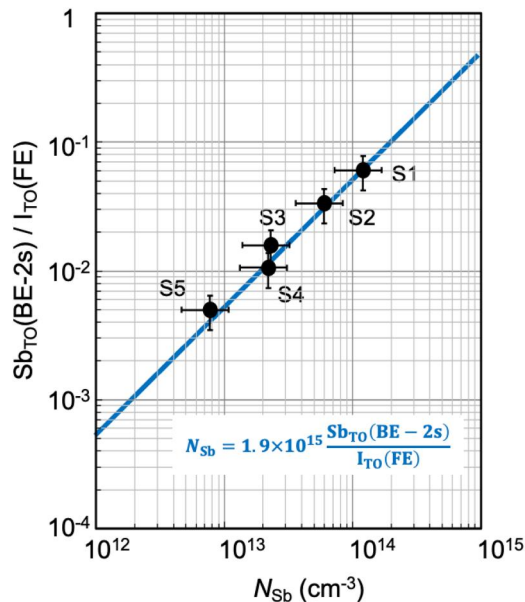


FIG. 4. Correlation between the Sb concentration, N_{Sb} , and the PL intensity ratio $\text{Sb}_{\text{TO}}(\text{BE-2s})/I_{\text{TO}}(\text{FE})$, used as a calibration curve for quantitative determination of Sb donors.

$$N_{\text{Sb}} = 1.9 \times 10^{15} \frac{\text{Sb}_{\text{TO}}(\text{BE} - 2\text{s})}{I_{\text{TO}}(\text{FE})}. \quad (3)$$

The data for sample S5, whose N_{Sb} was estimated using the standardized PL method after analytical separation, also lie on this line. We believe this relationship to be suitable as a calibration curve for determining N_{Sb} .

This linearity allows the applicability of this method to be extended to wide concentration ranges. Note that linear calibration curves have been obtained for B, P, Al, As, and Ga impurities in Si in the standardized PL method,^{13,15–22} where the ratios of BE to FE lines are used as measures of impurity concentrations. The detection limits of the BE and the FE lines determine the lower and upper limits of the applicable concentration range, respectively. This has been verified as stated in JIS.¹⁶ Since the BE2e transition is essentially derived from the BE transition, the linearity observed in the present calibration curve is a natural consequence. Therefore, we believe that the calibration curve can reasonably be extended in the same manner as the standardized PL method.

To verify the accuracy of this calibration curve, Sb concentrations for samples S1–S5 and C2 were determined using Eq. (3) and compared with concentrations obtained by the standardized PL method after analytical spectral separation.²⁷ The results are shown in Fig. 5, together with Sb concentrations measured by SIMS. As shown in the figure, the agreement among these methods is good, with discrepancies within approximately 30%, which is the same as the error specified in the standardized PL method.²¹ The fact that the Sb concentration could be determined in sample C2 codoped with Sb and P means that the present method is effective even when the concentrations of other coexisting impurities become high. This interference immunity to coexisting impurities has also been demonstrated in the standardized PL method: The quantitative analysis is possible even with samples that have a high compensation ratio¹⁶ and the quantified B concentration does not change before and after neutron transmutation doping.³⁴

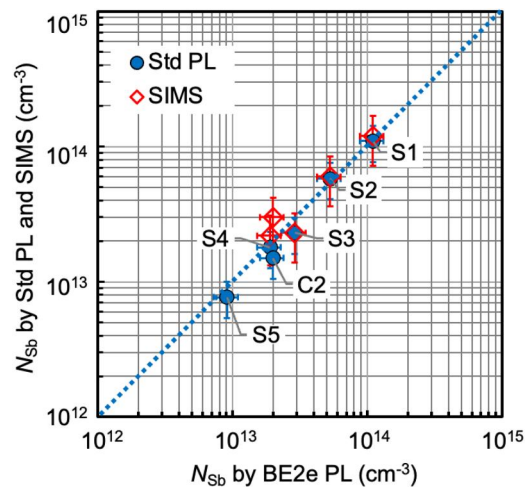


FIG. 5. Comparison of Sb concentrations obtained by the present BE2e PL method, those analytically derived from the standardized PL method (Std PL), and those measured by SIMS.

We investigated the error caused by the variation in the excitation power. Even when the excitation power was doubled, the change in the calibration curve was only approximately 5%. For reference, round-robin tests were performed in the standardized PL method to investigate changes in the intensity ratio of the BE to FE line due to variations in the excitation power, and the results showed that the change in the intensity ratio was within 20% for variations from half to twice the reference power.¹⁹ Therefore, the error due to the variation in the excitation power is within 30% of the overall error of the present method. These results indicate that the present method provides quantitative accuracy sufficient for practical applications.

Another characteristic feature of the present BE2e-transition-based method is its applicability to the quantification of residual P and B impurities in Sb-doped Si. The standardized PL method cannot directly quantify these impurities because the $P_{TO}(BE)$ and $B_{TO}(BE)$ lines spectrally coincide with the $Sb_{TO}(BE)$ and $Sb_{TO}(b_1')$ lines, respectively, making their separation impossible, as shown in Fig. 2. Consequently, in the standardized method, the $P_{TO}(BE)$ component had to be separated analytically, and the $B_{TO}(BE)$ component was estimated from the $B_{TO}(b_1)$ line in our previous report.²⁷

In contrast, in BE2e transition spectroscopy, the $P_{TO}(BE-2s)$ and $B_{TO}(BE-2\Gamma_g^+)$ lines are clearly separated from the $Sb_{TO}(BE-2s)$ line. If calibration curves similar to that shown in Fig. 4 for Sb impurities can be established for P and B impurities, direct quantification becomes feasible. This capability is demonstrated as follows: The ratios $P_{TO}(BE-2s)/I_{TO}(FE)$ and $B_{TO}(BE-2\Gamma_g^+)/I_{TO}(FE)$ exhibit small variations from sample S1 to S5 in Fig. 3, where the $I_{TO}(FE)$ line is not shown in the figure. Since these ratios serve as measures of the P and B concentrations, respectively, this result indicates that the concentrations of these impurities also vary only slightly across the samples. Preliminary experiments suggest that the magnitude of these variations is quantitatively consistent with that obtained by the standardized PL method.²⁷ We are currently extending the quantitative analysis based on BE2e transitions to typical donor and acceptor impurities B, P, Al, and As in Si.

A drawback of the present technique is that the intensity of the BE2e transition emission is approximately two orders of magnitude lower than that of the BE emission. Nevertheless, with modern PL instrumentation, high signal-to-noise ratio (S/N) detection is achievable under the conventional measurement conditions of the standardized PL method, and the detection sensitivity can be further improved by increasing the detector accumulation time to reduce random noise. Based on S/N of the PL spectrum of sample S5 in Fig. 3, the detection limit is estimated to be approximately $1 \times 10^{12} \text{ cm}^{-3}$ for a realistic accumulation time, which is only one order of magnitude higher than that of the standardized PL method using BE emission. The upper limit of quantification is determined by the detection limit of the $I_{TO}(FE)$ line, which decreases with increasing impurity concentration, and is therefore approximately $5 \times 10^{15} \text{ cm}^{-3}$, the same as the standardized PL method.²¹

In conclusion, we have demonstrated the effectiveness of quantitative analysis of Sb donors in Si using PL based on BE2e transitions. This technique enables spectral separation of Sb and P impurities that exhibit identical BE recombination energies. For Sb concentrations in the range from 7.7×10^{12} to $1.2 \times 10^{14} \text{ cm}^{-3}$, the intensity ratio of BE2e transition emission to FE emission is proportional to the Sb concentration, and this linear relationship can be used as a calibration

curve to quantify Sb concentrations over a wide range from 1×10^{12} to $5 \times 10^{15} \text{ cm}^{-3}$. The present BE2e PL method allows selective quantification of Sb donors even in the presence of residual P donors in Sb-doped high-resistivity MCZ Si, which has recently attracted attention for power-device applications. This technique is therefore expected to contribute significantly to improvements in the quality of high-resistivity Si crystals. Although PL based on BE2e transitions has rarely been applied to impurity characterization, this study demonstrates a novel and practical quantitative approach.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Michio Tajima: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (supporting); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (lead); Writing – review & editing (lead). **Yoshiji Miyamura:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (lead); Validation (supporting); Visualization (equal); Writing – original draft (supporting); Writing – review & editing (supporting). **Ryosuke Sakemi:** Data curation (supporting); Formal analysis (supporting); Investigation (supporting); Writing – review & editing (supporting). **Daigo Hayashi:** Data curation (supporting); Formal analysis (supporting); Investigation (supporting); Software (equal); Writing – review & editing (supporting). **Shin-ichi Nishizawa:** Conceptualization (supporting); Data curation (supporting); Formal analysis (equal); Funding acquisition (lead); Investigation (supporting); Methodology (supporting); Project administration (lead); Resources (lead); Software (supporting); Supervision (lead); Validation (equal); Visualization (supporting); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- ¹H. J. Shultze, H. Ofner, F. J. Niedernostheide, F. Lukermann, and A. Shultz, *IET Power Electron.* **12**, 3870 (2019).
- ²H. Minamitake, T. Yoshida, K. Suzuki, Y. Haraguchi, T. Hoshi, H. Koketsu, Y. Miyata, and A. Narazaki, in *Proceedings of 33rd International Symposium on Power Semiconductor Devices and ICs (ISPSD)* (IEEE 2021), p. 351.
- ³A. Abbadi, A. Desse, K. Melhem, S. D. Mariani, D. Fagiani, P. Zuliani, and F. Faller, *IEEE Trans. Semicond. Manuf.* **38**, 383 (2025).
- ⁴M. Shimoyama, in *Proceedings of 9th Meeting on Advanced Power Semiconductors* (The Japan Society of Applied Physics, 2022), p. 29 (in Japanese).

- ⁵M. Shimoyama, *Physics of Dopants Introduced into Silicon*, edited by Committee for Future Semiconductor Fund, The Japan Society of Applied Physics (Kindai Kagaku Digital Co, Tokyo, 2025), Chap. 11, p. 299 (in Japanese).
- ⁶S. E. Bkardshaw and A. I. Mlavsky, *J. Electron.* **2**, 134 (1956).
- ⁷W. Zulehner and D. Huber, *Growth, Properties and Applications*, Crystals Vol. 8, edited by J. Grabmaier (Springer-Verlag, Berlin, 1982), p. 26.
- ⁸J. Friedrich, W. von Ammon, and G. Müller, *Handbook of Crystal Growth* (Elsevier, 2015), p. 45.
- ⁹Z. Liu and T. Carlberg, *J. Electrochem. Soc.* **140**, 2052 (1993).
- ¹⁰M. Porrini, R. Scala, and V. V. Voronkov, *J. Cryst. Growth* **460**, 13 (2017).
- ¹¹T. Yatai, W. Saito, and S. Nishizawa, in *Ext. Abstr. SSDM* (The Japan Society of Applied Physics, 2025), p. 785.
- ¹²A. Kashizadeh, R. Basnet, A. Y. Liu, Z. Yang, L. Black, C. Sun, W. Han, Y. Wang, and D. Macdonald, *Sol. RRL* **9**, 2500335 (2025).
- ¹³M. Tajima, A. Yusa, and T. Abe, *Jpn. J. Appl. Phys., Part 1* **19**(Suppl. 19-1), 631 (1980).
- ¹⁴S. Nishizawa, *Physics of Dopants Introduced into Silicon*, edited by Committee for Future Semiconductor Fund, The Japan Society of Applied Physics (Kindai Kagaku Digital Co, Tokyo, 2025), Chap. 13, p. 343 (in Japanese).
- ¹⁵M. Tajima, *Appl. Phys. Lett.* **32**, 719 (1978).
- ¹⁶M. Tajima, *Semiconductor Technologies*, edited by J. Nishizawa (OHM, Tokyo, North-Holland, Amsterdam, 1981), p. 1.
- ¹⁷P. M. Colley and E. C. Lightowers, *Semicond. Sci. Technol.* **2**, 157 (1987).
- ¹⁸K. L. Schumacher and R. L. Whitney, *J. Electron. Mater.* **18**, 681 (1989).
- ¹⁹M. Tajima, T. Masui, D. Itoh, and T. Nishino, *J. Electrochem. Soc.* **137**, 3544 (1990).
- ²⁰I. Pelant and J. Valenta, *Luminescence Spectroscopy of Semiconductors* (Oxford, New York, 2012), p.190.
- ²¹Japanese Industrial Standards, Standard No. JIS H 0615:2021, 2021 (in Japanese).
- ²²Semiconductor Equipment and Materials International, Standard No. SEMI MF1389-1115, 2022.
- ²³E. C. Lightowers, M. O. Henry, and M. A. Vouk, *J. Phys. C* **10**, L713 (1977).
- ²⁴K. R. Elliott and T. C. McGill, *Solid State Commun.* **28**, 491 (1978).
- ²⁵A. S. Kaminskii, V. A. Karasyuk, and Y. E. Pokrovskii, *Sov. Phys. JETP* **47**, 1162 (1978); available at http://jetp.ras.ru/cgi-bin/dn/e_047_06_1162.pdf.
- ²⁶A. S. Kaminskii, L. I. Kolesnik, B. M. Leiferov, and Y. E. Pokrovskii, *J. Appl. Spectrosc.* **36**, 516 (1982).
- ²⁷Y. Miyamura, R. Sakemi, D. Hayashi, S. Nishizawa, and M. Tajima, *Jpn. J. Appl. Phys., Part 1* **64**, 12SP17 (2025).
- ²⁸P. J. Dean, J. D. Cuthbert, D. G. Thomas, and R. T. Lynch, *Phys. Rev. Lett.* **18**, 122 (1967).
- ²⁹P. J. Dean, J. R. Haynes, and W. F. Flood, *Phys. Rev.* **161**, 711 (1967).
- ³⁰R. Sauer, *J. Lumin.* **12/13**, 495 (1976).
- ³¹M. L. W. Thewalt, *Can. J. Phys.* **55**, 1463 (1977).
- ³²M. L. W. Thewalt, *Solid State Commun.* **23**, 733 (1977).
- ³³B. Pajot, *Optical Absorption of Impurities and Defects in Semiconducting Crystals: Hydrogen like Centres* (Springer, Heidelberg, 2010), Chap. 6 and 7, p. 169.
- ³⁴M. Tajima and A. Yusa, *Neutron Transmutation-Doped Silicon*, edited by J. Guldborg (Plenum, New York, 1981), p. 377.